

Brownian Motion

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Gaussian process I

Brownian motion is probably the most important continuous time stochastic process and has applications in a wide range of areas, such as finance, physics, engineering, and social science, to name a few. We present only the definition and some basic facts. We shall define Brownian motion as a special Gaussian process.

A **Gaussian process** X on T is a stochastic process such that for any $n \in \mathbb{N}$, $t_1, \dots, t_n \in T$, and $c_1, \dots, c_n \in \mathbb{R}$, the random variable $\sum_{i=1}^n c_i X_{t_i}$ is Gaussian/Normal. In other words, a stochastic process is Gaussian if its finite dimensional distributions are all Gaussian. A Gaussian process is said to be **centered** if $\mathbb{E}X_t = 0$ for all $t \in T$.

The following simple facts clarify the fundamental role played by the covariance function of a Gaussian process.

Theorem 1 (covariance function)

The distribution of a Gaussian process X on T is determined by the functions $\mathbb{E}X_t$ and $\text{cov}(X_s, X_t)$, $s, t \in T$.

Gaussian process II

Proof.

Let X and Y be two Gaussian processes on T with the same means and covariances. Then the random variables $\sum c_i X_{t_i}$ and $\sum c_i Y_{t_i}$ have the same mean and variance for any $n \in \mathbb{N}$, $c_1, \dots, c_n \in \mathbb{R}$ and $t_1, \dots, t_n \in T$. Therefore they have the same distribution since they are both Gaussian. This implies that $(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (Y_{t_1}, \dots, Y_{t_n})$ for all $t_1, \dots, t_n \in T$, $n \in \mathbb{N}$ (Here and hereafter, we use $\stackrel{d}{=}$ to mean equal in distribution.) Therefore $X \stackrel{d}{=} Y$ by Theorem 2.2 (Th. 2 from t2.pdf). □

Brownian motion I

A **(standard) Brownian motion** or **Wiener process** is a centered Gaussian process B in $\mathbb{R}$ defined on $\mathbb{R}_+$ with covariance function $\text{cov}(B_s, B_t) = s \wedge t$ and a.s. continuous sample paths $t \mapsto B_t$.

A stationary **Ornstein-Uhlenbeck (OU) process** is a centered Gaussian process on $\mathbb{R}_+$ with a.s. continuous sample paths and covariance function $\frac{1}{2}e^{-|t-s|}$, $s, t \geq 0$. The OU process has applications in such areas as term structure modelling and physics.

A **Brownian bridge** is a centered Gaussian process on $[0, 1]$ with a.s. continuous sample paths and covariance function $(s \wedge t)(1 - s \vee t)$, $s, t \geq 0$. The Brownian bridge has important applications in a number of areas including statistics (e.g. the Kolmogorov-Smirnov test and empirical process theory in general).

From the definition it is easy to see the following simple properties of the Brownian motion.

Brownian motion II

Proposition 2

Let B is a Brownian motion, and $\mathcal{F}_t = \sigma(\{B_s; s \leq t\})$. Then

- i. $B_0 = 0$ a.s.
- ii. $B_t \sim N(0, t)$, $t \geq 0$.
- iii. (Independent increments) For any $0 \leq s \leq t$, $B_t - B_s \perp\!\!\!\perp \mathcal{F}_s$.
- iv. (Stationary increments) For any $0 \leq s \leq t$, $B_t - B_s \stackrel{d}{=} B_{t-s}$.
- v. B_t is a martingale with respect to $\{\mathcal{F}_t\}$.
- vi. $-B_t$, $c^{-1}B_{c^2t}$, $c > 0$ are Brownian motions.
- vii. For any $u \geq 0$, $B_{u+t} - B_u$ is a Brownian motion.
- viii. $X_t = B_t - tB_1$, $t \in [0, 1]$ is a Brownian bridge.
- ix. $Y_t = e^{-t}B_{e^{2t}/2}$ is a stationary OU process.

Brownian motion III

Proof.

Note for any $n \in \mathbb{N}$ and $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$, $(B_{t_1}, \dots, B_{t_n})$ is Gaussian with zero mean and variance-covariance matrix

$$\begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_n \end{pmatrix}.$$

The claims then follow by straightforward calculations. □

Remark 3

*From Items iii and vii we see Brownian motion is **Markovian**, that is, for any $u \geq t$, X_u is independent of $\mathcal{F}_t$ given X_t . We will see later this property can be strengthened.*

Path property of Brownian motion I

The sample paths of the Brownian motion are quite peculiar continuous functions in that their quadratic variations in any interval is non-vanishing no matter how short the interval might be, as long as its length is not 0.

Proposition 4 (quadratic variation)

Let B be a Brownian motion, and fix any $t > 0$ and a sequence of partitions $0 = t_{n,0} < t_{n,1} < \dots < t_{n,k_n} = t$, $n \in \mathbb{N}$, such that $h_n = \max_k (t_{n,k} - t_{n,k-1}) \rightarrow 0$. Then

$$\zeta_n = \sum_k (B_{t_{n,k}} - B_{t_{n,k-1}})^2 \rightarrow t \text{ in } L^2. \quad (1)$$

Path property of Brownian motion II

Proof.

First note $B_t - B_s \stackrel{d}{=} |t - s|^{1/2} B_1$ and therefore

$$\mathbb{E}\zeta_n = \sum_k \mathbb{E}[(B_{t_{n,k}} - B_{t_{n,k-1}})^2] = \sum_k (t_{n,k} - t_{n,k-1}) \mathbb{E}B_1^2 = t.$$

Now (1) follows by noting

$$\begin{aligned} \mathbb{E}[|\zeta_n - t|^2] &= \text{var}(\zeta_n) = \sum_k \text{var}[(B_{t_{n,k}} - B_{t_{n,k-1}})^2] \\ &= \sum_k \text{var}(|t_{n,k} - t_{n,k-1}| B_1^2) \\ &= \sum_k (t_{n,k} - t_{n,k-1})^2 \text{var}(B_1^2) \\ &\leq h_n \sum_k (t_{n,k} - t_{n,k-1}) \text{var}(B_1^2) = h_n t \text{var}(B_1^2) \rightarrow 0. \end{aligned}$$



Path property of Brownian motion III

The last result implies that B has unbounded variation in any interval of positive length, which explains why the stochastic integral $\int_0^t f(s)dB(s)$ cannot be defined as an ordinary Stieltjes integrals (subject of your stochastic calculus course).

Corollary 5 (linear variation)

Brownian motion has a.s. unbounded variation on every interval $[s, t]$ with $s < t$.

Path property of Brownian motion IV

Proof.

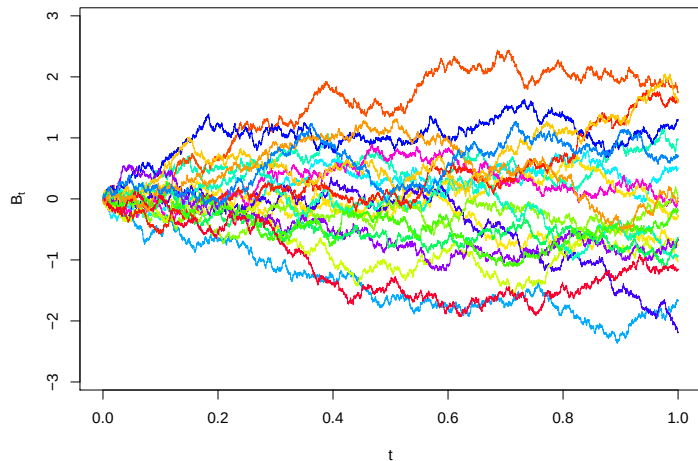
It suffices to consider the interval $[0, t]$, $t > 0$. If with positive probability the variation of B on $[0, t]$ is bounded, then there is a positive probability that

$$\sum_k (B_{t_{n,k}} - B_{t_{n,k-1}})^2 \leq \max_{k \leq k_n} |B_{t_{n,k}} - B_{t_{n,k-1}}| \sum_k |B_{t_{n,k}} - B_{t_{n,k-1}}| \rightarrow 0$$

for some sequence of partitions $\Pi_n : 0 = t_{n,0} < t_{n,1} < \dots < t_{n,k_n}$ s.t. $\|\Pi_n\| = \max_{k \leq k_n} |t_{n,k} - t_{n,k-1}| \rightarrow 0$. This contradicts Proposition 4. □

Brownian motion sample paths

See the following graph for a visual impression of the paths of a Brownian motion (up to time $t = 1$).



Strong Markov property I

Theorem 6 (strong Markov property)

Let B be a Brownian motion, $\mathcal{F}_t = \sigma \{B_s; s \leq t\}$, and $\mathcal{F}_t^+ = \mathcal{F}_{t+} = \bigcap_{h>0} \mathcal{F}_{t+h}$. For any a.s. finite $\{\mathcal{F}_t^+\}$ -stopping time τ , the process $B'_t = B_{\tau+t} - B_\tau$, $t \geq 0$ is again a Brownian motion independent of $\mathcal{F}_\tau^+$.

Strong Markov property II

Proof.

First note that $B_{u+t} - B_u \perp\!\!\!\perp \mathcal{F}_{u+}$, $t \geq 0$, since for all $h > 0$, $B_{u+h+t} - B_{u+h} \perp\!\!\!\perp \mathcal{F}_{u+h} \supset \mathcal{F}_{t+}$ and $B_{u+t} - B_u = \lim_{h \downarrow 0} (B_{u+h+t} - B_{u+h})$. Let $\tau_n = 2^{-n}[2^n\tau + 1]$. Then the τ_n are stopping times that take countably many values and $\tau_n \downarrow \tau$ as $n \rightarrow \infty$. Let $B_t^n = B_{\tau_n+t} - B_{\tau_n}$, $t \geq 0$. Now fix any $A \in \mathcal{B}(C[0, \infty))$ and $E \in \mathcal{F}_{\tau_n}^+$,

$$\begin{aligned} P(\{B^n \in A\} \cap E) &= \sum_k P(\{B^n \in A\} \cap E \cap \{\tau_n = k2^{-n}\}) \\ &= \sum_k P(\{B_{k2^{-n}+} - B_{k2^{-n}} \in A\} \cap E \cap \{\tau_n = k2^{-n}\}) \\ &= \sum_k P(\{B_{k2^{-n}+} - B_{k2^{-n}} \in A\})P(E \cap \{\tau_n = k2^{-n}\}) \\ &= \sum_k P(\{B \in A\})P(E \cap \{\tau_n = k2^{-n}\}) \\ &= P(\{B \in A\})P(E). \end{aligned}$$

Therefore for all n , $B^n \perp\!\!\!\perp \mathcal{F}_{\tau_n}^+$ and B^n is a Brownian motion.

Strong Markov property III

Proof.

The continuity of B yields $B_t^n \rightarrow B'_t$ a.s. for every $t \geq 0$. By dominated convergence we then obtain for any $E \in \mathcal{F}_\tau^+ \subset \mathcal{F}_{\tau_n}^+$ and $t_1, \dots, t_k, k \in \mathbb{N}$, and for any bounded continuous function $f : \mathbb{R}^k \rightarrow \mathbb{R}$,

$$\begin{aligned}\mathbb{E}[f(B'_{t_1}, \dots, B'_{t_k}); E] &= \lim_n \mathbb{E}[f(B_{t_1}^n, \dots, B_{t_k}^n); E] \\ &= \lim_n \mathbb{E}[f(B_{t_1}^n, \dots, B_{t_k}^n)]P(E) = \mathbb{E}[f(B_{t_1}, \dots, B_{t_k})]P(E)\end{aligned}$$

By approximating the indicator functions $1_{A_1 \times \dots \times A_k}$ of closed rectangles using an increasing sequence of bounded continuous functions and by monotone convergence, we have $P(\{B' \in \cdot\} \cap E) = P(\{B \in \cdot\})P(E)$ hold for cylinder sets. Then by a monotone class argument, we have $P(\{B' \in \cdot\} \cap E) = P(\{B \in \cdot\})P(E)$ for general measurable sets. $\square$

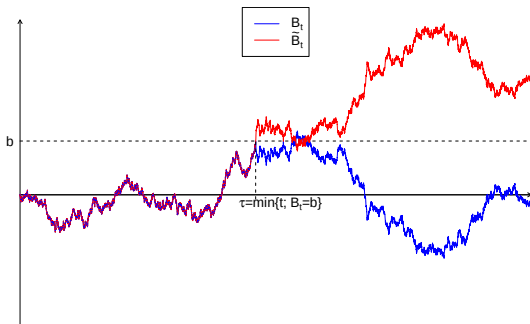
Proposition 7 (reflection principle)

Let B be a Brownian motion and τ an associated stopping time. Then the reflected process $\tilde{B}$ defined by

$$\tilde{B}_t = B_{t \wedge \tau} - (B_t - B_{t \wedge \tau}), \quad t \geq 0$$

is also a Brownian motion.

The following figure illustrates the reflected process when $\tau = \inf \{t; B_t = b\}$.



Proof.

It is enough to compare the distributions of B and $\tilde{B}$ up to a fixed time t , and so we may assume $\tau < \infty$. Let $B_t^\tau = B_{\tau \wedge t}$ and $B'_t = B_{\tau+t} - B_\tau$. By Theorem 6, $B' \perp\!\!\!\perp (\tau, B^\tau)$. Moreover, by $-B' \stackrel{d}{=} B'$, we have $(\tau, B^\tau, B') \stackrel{d}{=} (\tau, B^\tau, -B')$. The result now follows from

$$B_t = B_t^\tau + B'_{(t-\tau)_+}, \quad \tilde{B}_t = B_t^\tau - B'_{(t-\tau)_+}, \quad t \geq 0. \quad \square$$

Maximum process I

Proposition 8 (maximum process)

Let B be a Brownian motion, and define $M_t = \sup_{s \leq t} B_s$, $t \geq 0$.

Then

$$M_t \stackrel{d}{=} M_t - B_t \stackrel{d}{=} |B_t|, \quad t \geq 0.$$

Maximum process II

Proof.

We only need to consider the case $t > 0$. By scaling, it further reduces to the case $t = 1$. We now prove by calculating and comparing the density functions. First find the density of (M_1, B_1) . For $x \geq 0$ and $y \leq x$, applying Proposition 7 with $\tau = \min \{t; B_t = x\}$ yields

$$P \{M_1 \geq x, B_1 \leq y\} = P \left\{ \tilde{B}_1 \geq 2x - y \right\} = P \{B_1 \geq 2x - y\}.$$

By differentiation the density of (M_1, B_1) is $-2\varphi'(2x - y)$, where φ is the standard normal density. Changing variables, we may conclude that $(M_1, M_1 - B_1)$ has density $-2\varphi'(x + y)$, $x, y \geq 0$. In particular, the density of M_1 and $M_1 - B_1$ are both $2\varphi(x)$, $x \geq 0$, which clearly is also the density of $|B_1|$. □

Itô's integral I

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ be a filtered probability space. Let $\{B_t, t \geq 0\}$ be a standard Brownian motion adapted to $\{\mathcal{F}_t\}_{t \geq 0}$.

Let $\mathcal{P}$ denote the smallest σ -algebra of subsets of $[0, \infty) \times \Omega$ such that all left continuous $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted processes $Y : [0, \infty) \times \Omega \mapsto \mathbb{R}$ are measurable. A process $X = \{X_t, t \geq 0\}$ is called **predictable** if it is measurable w.r.t. $\mathcal{P}$.

We shall define the stochastic integral $\int_0^t f_s dB_s$, $0 \leq t \leq T$ for predictable processes f such that $\int_0^T f_s^2 ds < \infty$ a.s.

A process f is said to be simple predictable if $f_t = \xi_{k-1}$ for $(k-1)2^{-n} < t \leq k2^{-n} \leq T$ and ξ_{k-1} is $\mathcal{F}_{(k-1)2^{-n}}$ -measurable for each k .

Itô's integral II

For a simple predictable process, define $I(f)_t := \int_0^t f_s dB_s$ by

$$\begin{aligned} I(f)_t &= \sum_{k=1}^{[2^n t]} \xi_{k-1} (B_{k2^{-n}} - B_{(k-1)2^{-n}}) + \xi_{[2^n t]} (B_t - B_{[2^n t]2^{-n}}). \\ &= \sum_{k=1}^{\infty} \xi_{k-1} (B_{t \wedge k2^{-n}} - B_{t \wedge (k-1)2^{-n}}). \end{aligned}$$

For a predictable process f s.t. $\int_0^T f(s)^2 ds < \infty$ a.s. choose a sequence of simple predictable processes $\{f_n\}$ s.t.

$$\int_0^T (f(s) - f_n(s))^2 ds \leq 2^{-n} \text{ for all sufficiently large } n, \text{ a.s.,}$$

and then define the Itô integral $I(f)(t) = \int_0^t f_s dB_s$ as,

$$\int_0^t f_s dB_s = \lim_{n \rightarrow \infty} \int_0^t f_n(s) dB(s).$$

Itô's integral III

The Itô integral has the following properties:

(a) (Linearity)

$$\int_0^t \alpha f_1(s) + \beta f_2(s) dB(s) = \alpha \int_0^t f_1(s) dB(s) + \beta \int_0^t f_2(s) dB(s),$$

$\alpha, \beta \in \mathbb{R}.$

(b) If $\|f\|^2 = \mathbb{E}[\int_0^\infty f^2(s) ds] < \infty$, then

$$\mathbb{E} \left[\left(\int_0^\infty f(s) dB(s) \right)^2 \right] = \mathbb{E} \left[\int_0^\infty f^2(s) ds \right], \quad (\text{Itô's isometry.})$$

and $\int_0^t f(s) dB(s)$, $t \geq 0$ is a martingale.

Itô's integral IV

Lemma 9

Let $z_n(t) = \sum_{k=1}^{\lfloor t2^n \rfloor} (B_{k/2^n} - B_{(k-1)/2^n})^2 + (B_t - B_{\lfloor t2^n \rfloor/2^n})^2 - t$.

Then

$$P \left\{ \max_{t \leq T} |z_n(t)| < n2^{-n/2}, \text{ for all sufficiently large } n \right\} = 1.$$

Itô's integral V

Proof.

First note $\mathbb{E}[z_n(t)^2] \sim 2t2^{-n}$ by some algebra. Then note that $z(t)$ is a martingale, and so by Doob's inequality,

$$\begin{aligned} P \left\{ \max_{t \leq T} |z_n(t)| \geq n2^{-n/2} \right\} &= P \left\{ \max_{t \leq T} z_n(t)^2 \geq n^2 2^{-n} \right\} \\ &\leq n^{-2} 2^n \mathbb{E}[z_n(T)^2] \\ &\leq Cn^{-2}. \end{aligned}$$

So, $\sum_n P \left\{ \max_{t \leq T} |z_n(t)| \geq n2^{-n/2} \right\} < \infty$, and therefore $P \left\{ \max_{t \leq T} |z_n(t)| \geq n2^{-n/2} \text{ infinitely often} \right\} = 0$. □

Itô's integral VI

Theorem 10

$$\int_0^t B_s dB_s = \frac{1}{2} B_t^2 - \frac{1}{2} t.$$

Proof.

For any n , let $t_k^n = k2^{-n}$, for $k \leq [2^n t]$; $= t$ for $k \geq [2^n t] + 1$. Note

$$B_t^2 = \sum_{k \geq 1} B_{t_k^n}^2 - B_{t_{k-1}^n}^2 = \sum_{k \geq 1} (B_{t_k^n} - B_{t_{k-1}^n})^2 + 2 \sum_{k \geq 1} B_{t_{k-1}^n} (B_{t_k^n} - B_{t_{k-1}^n}),$$

and so

$$\sum_{k \geq 1} B_{t_{k-1}^n} (B_{t_k^n} - B_{t_{k-1}^n}) = \frac{1}{2} B_t^2 - \frac{1}{2} \sum_k (B_{t_k^n} - B_{t_{k-1}^n})^2.$$

Taking limit on both sides, the RHS converges to $\frac{1}{2} B_t^2 - \frac{1}{2} t$ by the previous lemma, while the LHS converges to $\int_0^t B_s dB_s$. $\square$

Brownian semimartingales and Itô's Lemma I

A **Brownian semimartingale** is a process

$$X(t) = X(0) + \int_0^t f(s)dB(s) + \int_0^t g(s)ds, \quad (2)$$

$$[dX(t) = f(t)dB(t) + g(t)dt, \text{ for short }] \quad (3)$$

for an $\mathcal{F}_0$ -measurable $X(0)$, a predictable process f with $\int_0^t f^2(s)ds < \infty$, $\forall t \geq 0$ a.s., and a predictable process g with $\int_0^t |g(s)|ds < \infty$, $\forall t \geq 0$ a.s..

Brownian semimartingales and Itô's Lemma II

Lemma 11 (Itô's Lemma)

Let $u : [0, \infty) \times \mathbb{R}^d \mapsto \mathbb{R} : (t, x_1, \dots, x_d) \mapsto u(t, x_1, \dots, x_d)$ be a function with $u_0 = \partial u / \partial t$, $u_j = \partial u / \partial x_j$, $u_{ij} = \partial^2 u / \partial x_i \partial x_j$, $i, j = 1, \dots, d$, all continuous. If $X_1, \dots, X_d$ are Brownian semimartingales, then $u(t) = u(t, X_1(t), \dots, X_d(t))$ is also a Brownian semimartingale with the representation

$$\begin{aligned} du(t) &= u_0(t, X_1(t), \dots, X_d(t))dt \\ &\quad + \sum_{j=1}^d u_j(t, X_1(t), \dots, X_d(t))dX_j(t) \\ &\quad + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d u_{ij}(t, X_1(t), \dots, X_d(t))dX_i(t)dX_j(t), \end{aligned}$$

where the $dX_i(t)dX_j(t)$ are calculated by the following rules, $(dt)^2 = 0$, $dt dB(t) = 0$, $(dB(t))^2 = dt$.

Brownian semimartingales and Itô's Lemma III

Example 12

- 1) $dB^2(t) = 2B(t)dB(t) + (dB(t))^2 = 2B(t)dB(t) + dt$, which also gives $\int_0^t B(t)dB(t) = \frac{1}{2}B^2(t) - \frac{1}{2}t$.
- 2) If $z(t)$ is such that $dz(t) = z(t)f(t)dB(t)$, with $z(0) = 1$, then by applying Itô's Lemma on $\log z(t)$, we get
$$z(t) = \exp \left\{ \int_0^t f(s)dB(s) - \frac{1}{2} \int_0^t f(s)^2 ds \right\}.$$